

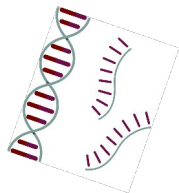
Introduction to Nanopore (ONT) sequencing

ONT mNGS pipeline

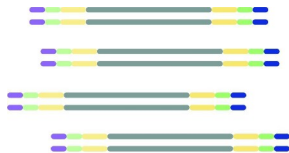
Study
design



xNA
extraction



Library prep



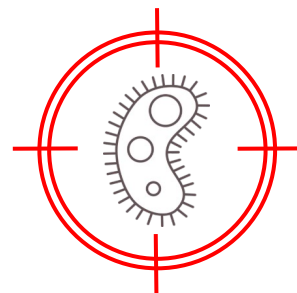
Sequencing



Data analysis



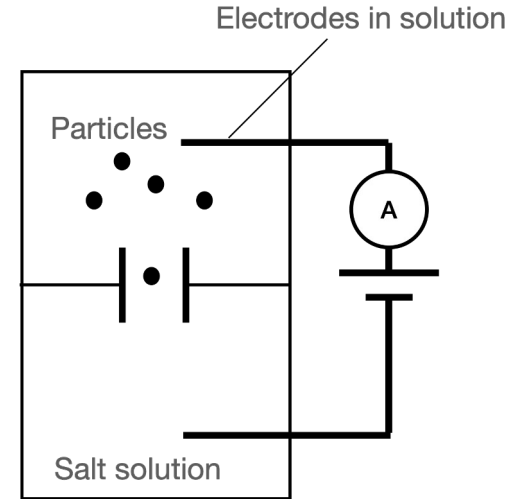
Pathogen ID



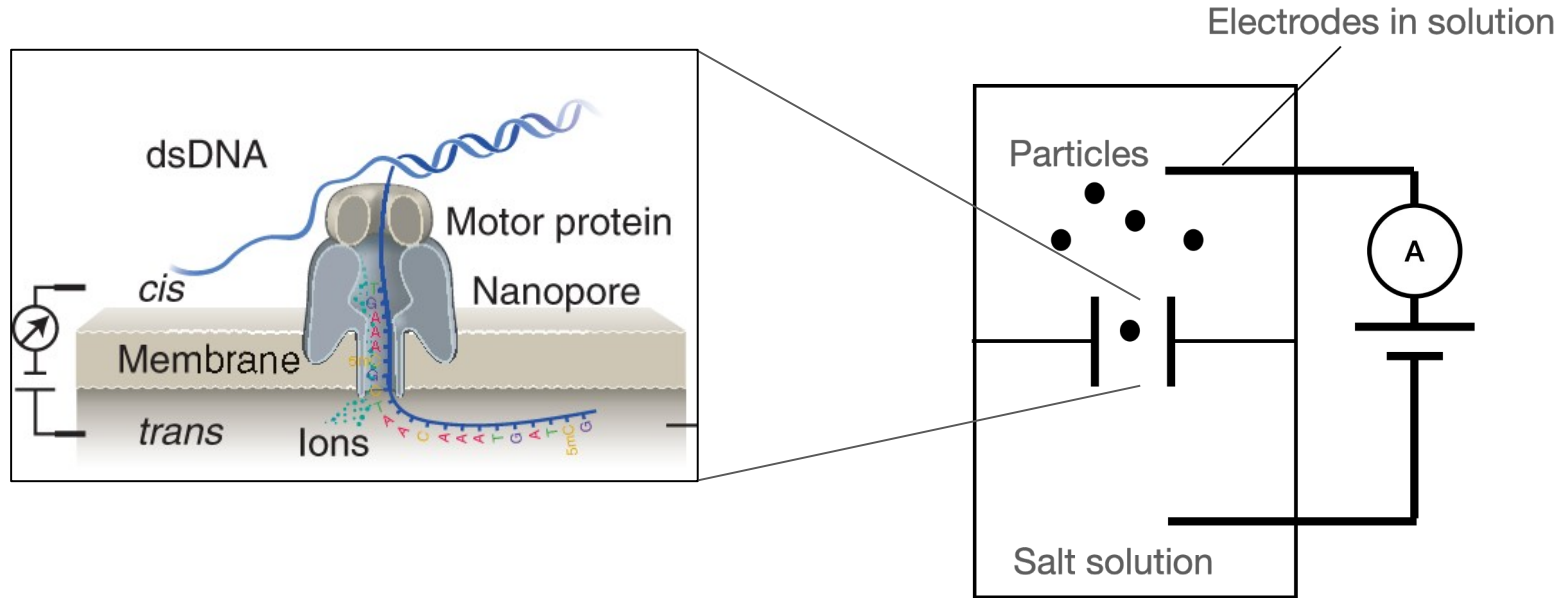
INTRODUCTION

What is nanopore sequencing?

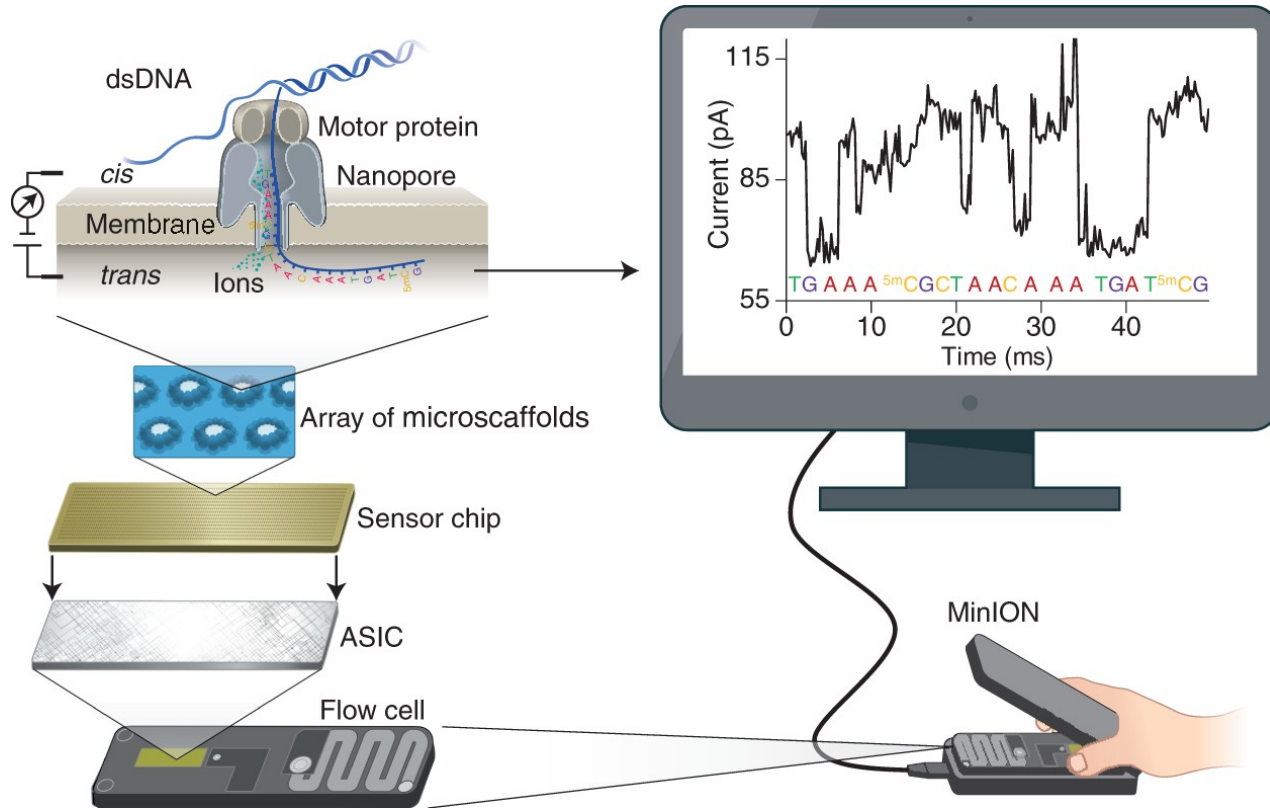
- Use of nano-scale sensors to detect sequence of nucleotides
- Sensors work by detecting changes in electrical current as DNA/RNA particles flow through a hole (“pore”)
- ONT is one of two commercially available nanopore platforms (other is Qitan)



Principles of Oxford Nanopore sequencing

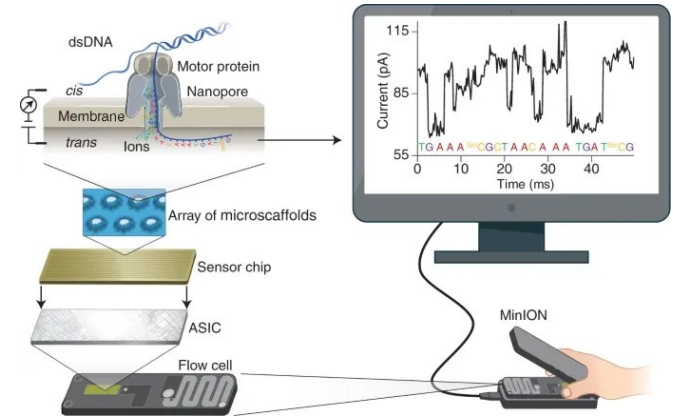


Principles of Oxford Nanopore sequencing



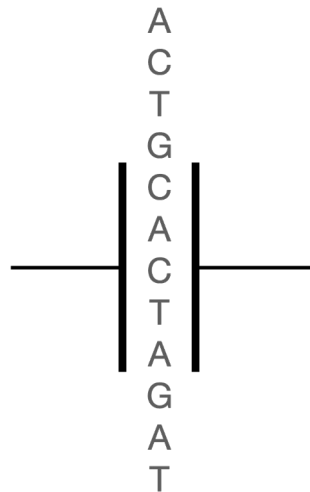
Principles of Oxford Nanopore sequencing

- Biological pores and motor proteins used to guide DNA/RNA across electrical gradient
- Changes in electrical signal during this passage (due to blocking of electrons) detected electronically



Principles of Oxford Nanopore sequencing

- Size of the pore accommodates 5 nucleotides
- DNA/RNA translocation typically happens at ~million bases/second, which is too fast
- Translocation is slowed down to ~1000 bases/second in ONT to allow for discrimination of signals



Different ONT instruments

NANOPORE PRODUCT FAMILY

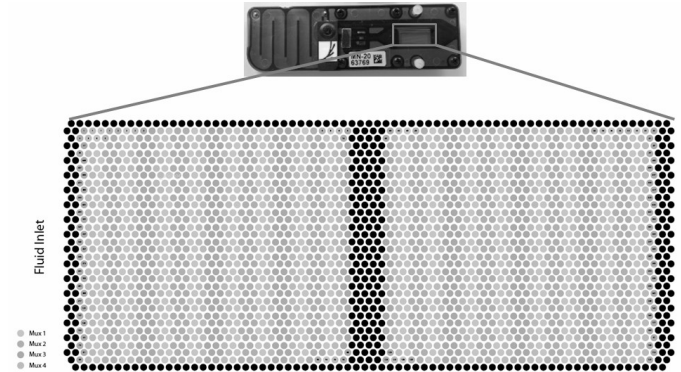
One core technology, real-time, on-demand



Key	SmidgION	Flongle	MinION	GridION	PromethION
System Price	TBC	Included in \$5K Starter Pack	Included in \$1K Starter Pack	Included in \$50K Starter Pack	Included in \$135K Starter Pack
Number of channels	200 channels	128 channels	512 channels	5 x 512 = 2,560*	48 x 3,000* = 144,000
Per flow cell Current Data – Max Data	TBC	1 - 3.3 Gb	17 - 40 Gb	17 - 40 Gb	125 - 311 Gb
Per Device Current Data – Max Data				85 - 200 Gb	3/6 - 20 Tb
Price per Gb Current Data – Max Data	TBC	\$90 - \$30	\$30 - \$12.5	\$17.5 - \$7.5	\$5 - \$2

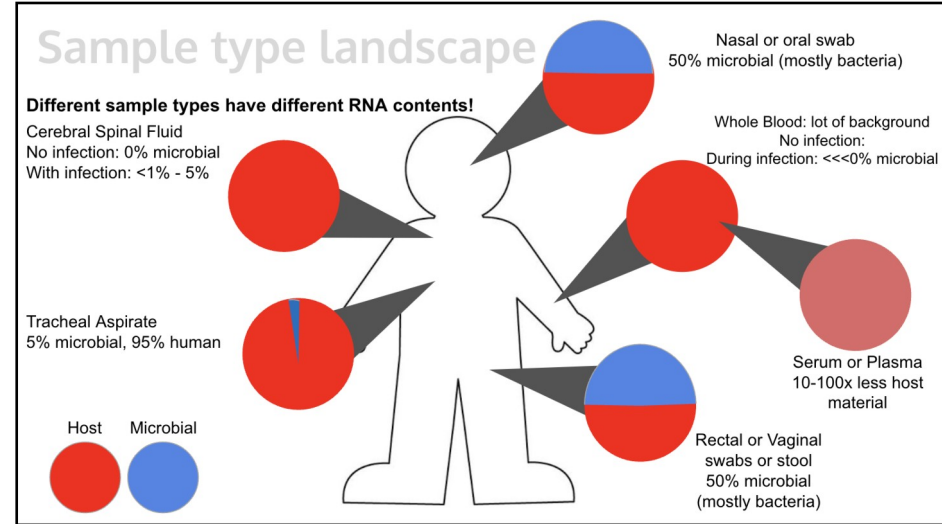
The MinION flow cell (see [here](#) for more)

- Each flow cell has 2048 pores
- Typically only ~1200-1600 are actually available on a new flow cell
- Signal from 512 pores can be detected simultaneously



Considerations when doing metagenomics (mNGS)

- DNA or RNA mNGS?
 - Purpose of study
 - Pathogen landscape
- Sample type
 - Total nucleic acid abundance
 - Proportion of host vs. microbe



Considerations when doing Nanopore (ONT) sequencing

- Sample integrity
 - Important to ensure high quality with little to no degradation
 - True for Illumina as well, but even more so for ONT
- Library preparation kit ([link](#) to available kits)
- DNA or RNA?
- PCR or PCR-free?
- Multiplexing or single sample?

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Current kit of choice: Rapid PCR barcoding kit (SQK-RPB114.24)